

Shuo Wu

Ph.D. Candidate (ECE) / Machine Learning Engineering / LLM Systems: Evaluation, Efficiency, Reproducible Benchmarking

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Summary

- ML engineering + research background: build **evaluation harnesses**, **metrics/logging**, and **data preprocessing pipelines** for LLM/RL systems with reproducible benchmarking and ablation reporting.
- Delivered measurable efficiency gains: **15–35% token reduction** at equal-or-better accuracy in a teacher–student multi-LLM cascade; **~20% fewer environment interactions** in hybrid model-based/model-free optimization.
- Strong mathematical modeling and optimization: implemented **MILP** formulations and **solver-backed pipelines** (Gurobi) for robust reward design and empirical validation.

Experience / Projects

LLM Systems Research Collaboration, Rutgers University–New Brunswick (joint project) Oct 2024–Aug 2025

Interactive LLM Cascade (Teacher–Student): LLM routing, evaluation harness, cost/latency

- Built an **online interactive LLM cascade** where a strong model distills reusable **problem-solving strategies** and injects them into future prompts to improve a weak model **without fine-tuning**.
- Improved weak-model accuracy (up to +**33.06** points) and overall system accuracy (up to +**5.53** points) while reducing strong-model usage (up to **48.05% fewer** calls) and fees (up to **49.63%**).
- Engineered a reproducible **evaluation harness** (dataset loaders, batched inference, metrics, ablations/error slicing) with **schema-validated JSON outputs** and **tool/function calling** for reliable logging and automated post-processing.

RAG Faithfulness Research, University of Illinois Chicago

Mar 2025–Present

Attention-Guided Context Selection for Faithful RAG Generation LLM internals, retrieval, hallucination

- Investigating whether **prefill-stage attention scores** (query→chunk) can serve as a pre-generation signal to rerank retrieved chunks and improve **answer faithfulness** in RAG systems.
- Designed a correlation analysis pipeline to evaluate the relationship between attention-based chunk scores and faithfulness labels on **HaluEval**, benchmarking against BM25 and dense retrieval baselines.
- Situating the approach relative to post-generation detection methods (Lookback Lens, ReDeEP) and efficiency-focused pruning (AttentionRAG); targeting a mechanistic account of **why** certain chunks elicit faithful generation.

Hybrid RL Research Collaboration, Rutgers University–New Brunswick (joint project) Jul 2024–Aug 2024

Data Filtering for Dyna-Style Model-Based RL MBPO, OOD detection, experiment automation

- Implemented an **OOD filter** that rejects out-of-distribution model rollouts before policy updates, improving synthetic-data quality in Dyna-style training.
- Integrated into **MBPO** and built an experiment pipeline (sweeps, logging, curve aggregation) to evaluate sample-efficiency and final-return across **MuJoCo** tasks.

Technical Skills

Languages: Python, MATLAB, R

ML/LLM: PyTorch, NumPy, scikit-learn, FAISS; LLM routing/cascades; benchmarking and ablation

studies

Engineering: data preprocessing scripts; evaluation harnesses; metrics design; structured logging; reproducible experiment automation

Optimization: Gurobi; ILP/MILP modeling; MDPs

Tools: Git/GitHub, Linux, Jupyter, LaTeX

Familiar with: unit testing (pytest), CI (GitHub Actions)

Additional Engineering Projects

- **RAG Prototype for Benchmarking:** Implemented embedding + FAISS retrieval for evaluation workflows; scripted end-to-end runs and metric aggregation.
- **Logic-to-Solver Data Pipeline:** Transformed FOL $\rightarrow$ AST $\rightarrow$ Z3 constraints to support automatic verification and controlled perturbations for reasoning tasks.

Selected Publications / Preprints

(* = equal contribution)

- **Not only a helper, but also a teacher: Interactive LLM Cascade.** *Submitted to ICML 2026 (under review).* *arXiv preprint*, 2025. [arXiv:2509.22984](#).
Yu Wu*, [Shuo Wu](#)*, Ye Tao*, Yansong Li, Anand D. Sarwate.
- **When to Trust Your Data: Enhancing Dyna-Style Model-Based Reinforcement Learning With Data Filter.** *arXiv preprint*, 2024. [arXiv:2410.12160](#).
Yansong Li, Zeyu Dong, Ertai Luo, Yu Wu, [Shuo Wu](#), Shuo Han.
- **Robust Reward Design for Markov Decision Processes.** *Journal of Artificial Intelligence Research (JAIR)*, 2025.
[Shuo Wu](#), Haoxiang Ma, Jie Fu, Shuo Han.
- **Margin-Based Sparsification for Feature-Budgeted Linear SVM.** *Submitted to ISAIM 2026*.
Lev Reyzin, [Shuo Wu](#).
- **Margin-Based Boosting for Partial Interpretability.** *Submitted to ICML 2026 (under review)*.
Xing Gao, Lev Reyzin, [Shuo Wu](#).

Education

University of Illinois Chicago

Ph.D., Electrical and Computer Engineering

Advisor: Prof. Shuo Han, Prof. Lev Reyzin

Southern University of Science and Technology

B.S., Mathematics and Applied Mathematics

Chicago, IL

Sep 2022–Jul 2027 (expected)

Shenzhen, China

Sep 2018–Jul 2022